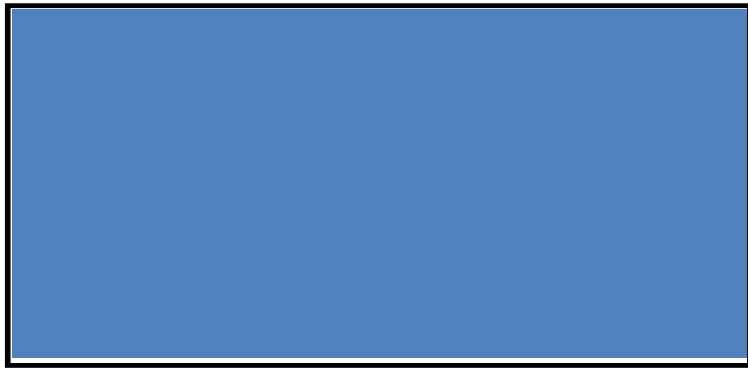
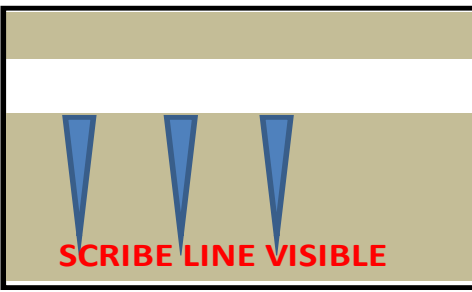
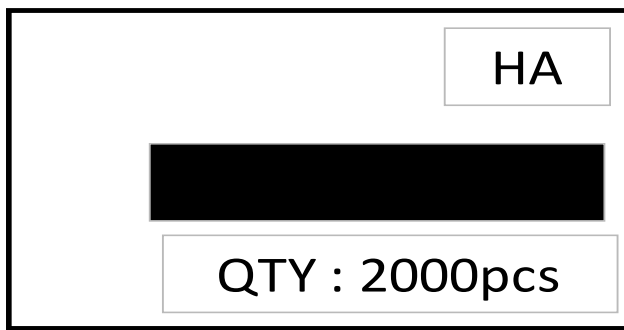
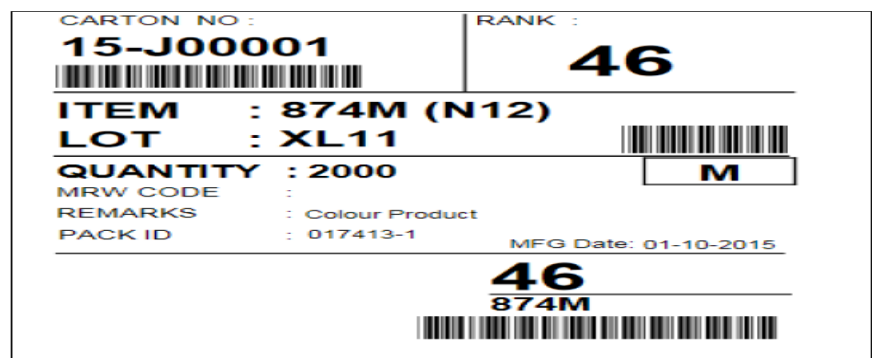


Customer: SAMSUNG TIANJIN		Date: 12-Oct-15	Time: 14:00
Participant From Maruwa		Participant From Organizer	Report To
1. Sivakumar	QA	1. Mr Ohsung Kwon	Principal Eng
2. Mr Izumo	MM Eng	2. Mr Cui Chengzhe	Purchasing AM
3. Ms Lu	Korea MW	3. Ms Dong Jingwen	QA Asst Mgr
4.		4.	
5.		5.	

Agenda

Meeting started of with Quality group.
The details as below (Please refer to the attach sheet for details)

SAMSUNG QUALITY MEETING						12/10/2015
No	Complaint	Product	Image	Samsung Comment	Maruwa Action	PIC
1	Laser scribe NG (This claim is very high priority)	65 x 74 x 0.18T	AFTER SCRIBBING  MARUWA PRODUCT  NO SCRIBE LINE VISIBLE T PRODUCT  SCRIBE LINE VISIBLE	Samsung will submit report for the 5 evaluated sample. Samsung still believe Maruwa need to look into the material content . The sample evaluated on the B3 temp to B7 temp RESULT: FAIL	Sample submitted and evaluated by Samsung Result not satisfactory and Maruwa will re-submit sample for evaluation Criteria: a. Easy to break b. No crack during laser scribe	Mr Siva Mr Izumo
2	Audit finding	All process	-	Closed all the finding in Sept 2014	Submitted all the all the audit findig with evidence.	Mr Siva
3	Carton label	65X74x0.18T		SANSUNG have requested MARUWA to include the indication of HA/HO and MARUWA said it can't be done but recents shipments, the HA/HO indication have been indicated in the label. If this can be done, Samsung wants MARUWA to change the specification	MARUWA will check and arrange to change the specification	Mr Izumo
5	Carton label	All product		Request to submit ECN for outer label.	Maruwa propose to add in the MFG DATE in the outer label	Mr Siva

Other discussion/feedback

- Maruwa's sample still fail comparison with ECT's sample. Samsung is looking at few items such as
 - Breakage strenght => Prefer product easy to break
 - No crack after laser => profile image
 - Request Maruwa to submit sample report before shipping out sample. Explained that Maruwa need to identify the cause immediately or else all the stock will be NG.
 - Comment on the order status

RN104PA => Reducing

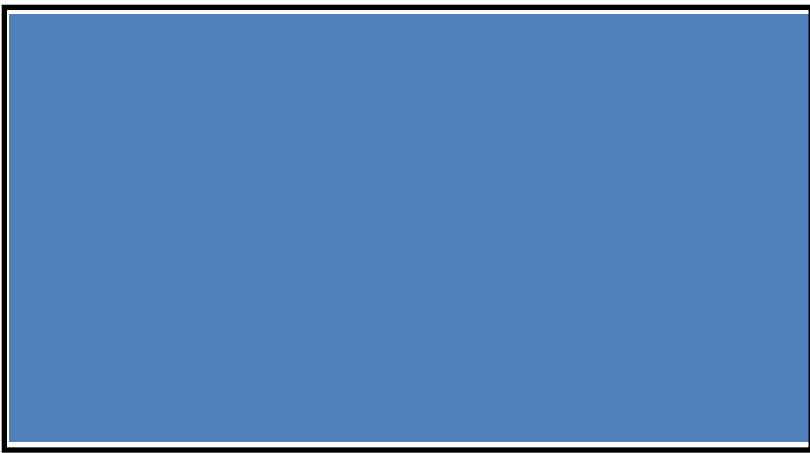
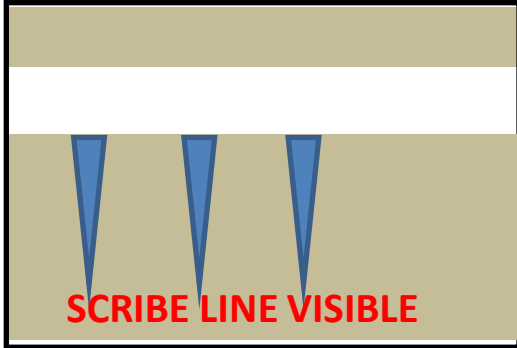
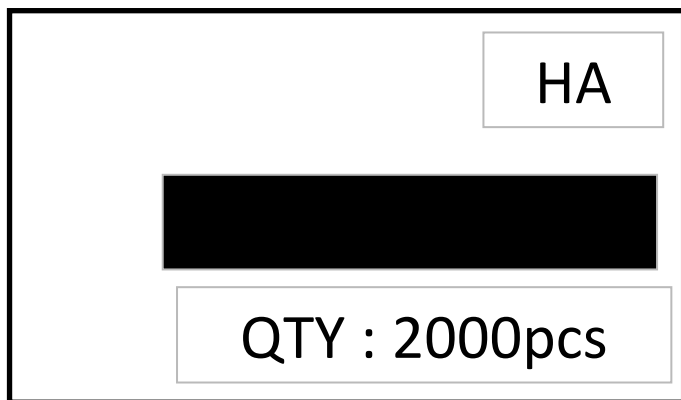
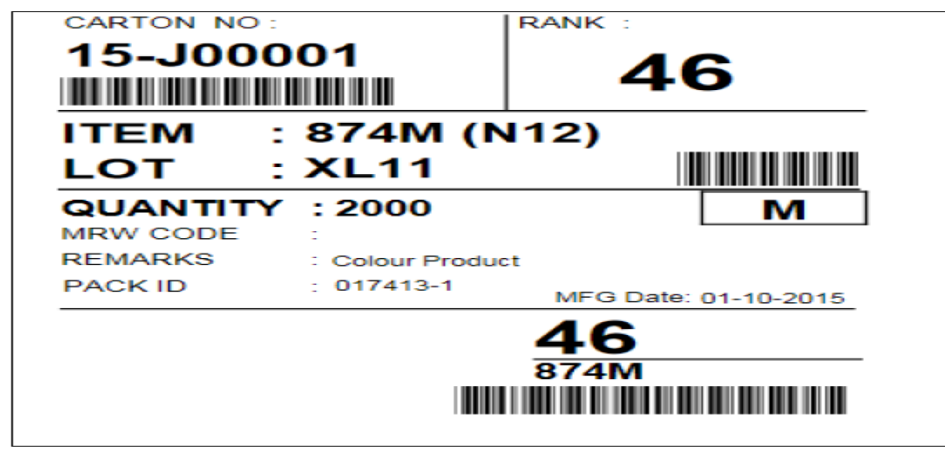
RN102P => Increasing

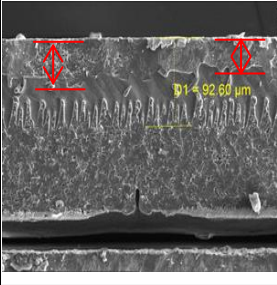
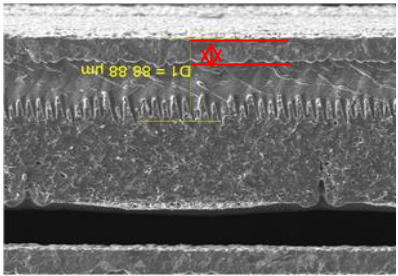
69x49.5x0.1T => Quantity increasing because this product used in SAMSUNG WATCH
 - Maruwa need to submit 65x74x0.4T specification (pending for some time) => Mr Izumo
- Meeting adjourned at 3:45pm

Conclusion & Opinion

Comment from Principal Engineer : Mr Ohsung Kwon

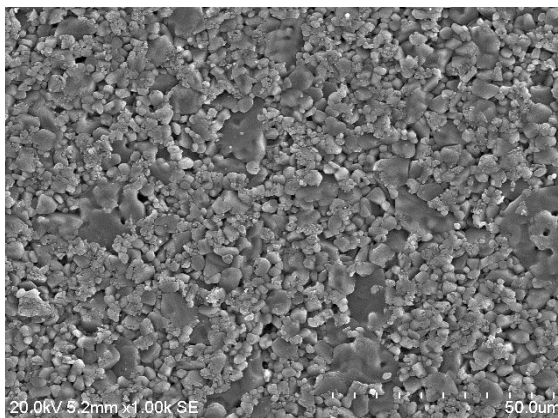
He understand the current quality issue and Samsung would also want support Maruwa by issuing PO's but for this 0.18t, Maruwa need to expeditite the activity since this failure is very critical and it is Samsuf core production.

No	Complaint	Product	Image	Samsung Comment	Maruwa Action	PIC
1	Laser scribe NG (This claim is very high priority)	65 x 74 x 0.18T	AFTER SCRIBBING  MARUWA PRODUCT  T PRODUCT 	Samsung will submit report for the 5 evaluated sample. Samsung still believe Maruwa need to look into the material content . The sample evaluated on the B3 temp to B7 temp RESULT: FAIL	Sample submitted and evaluated by Samsung Result not satisfactory and Maruwa will re-submit sample for evaluation Criteria: a. Easy to break b. No crack during laser scribe	Mr Siva Mr Izumo
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区分	65 x74 x 0.18T	
	MA-未分解	T社
SEM		

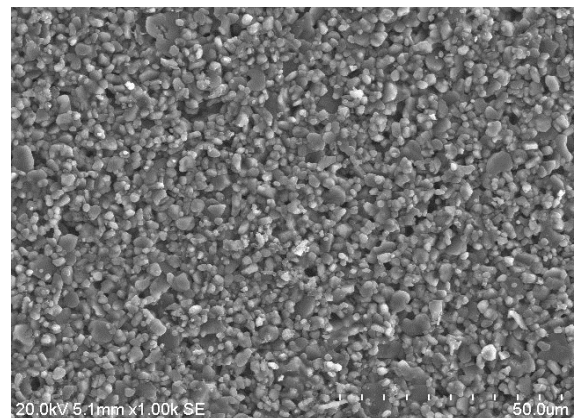
Analysis Sample MW & TC

MARUWA



JUDGEMENT: BAD

SAMPLE TC



GOOD

Potential Cause

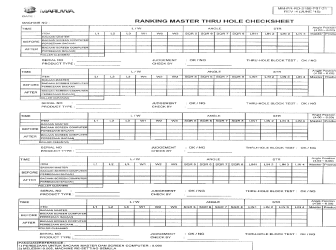
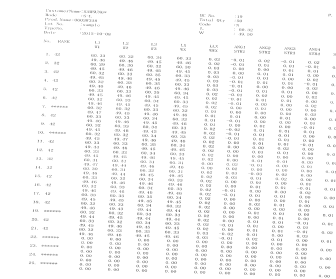
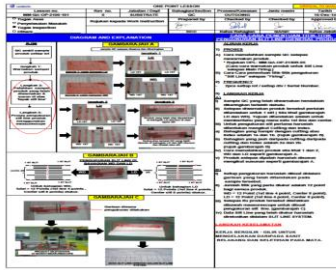
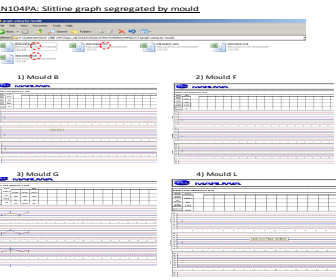
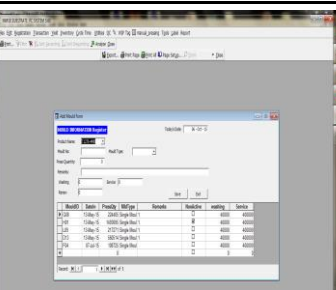
Due to particle size is big, the formation is big which causes breakage issue after laser

Action

Current particle spec maintain at standard spec and decided to change according to product type

SAMPLE submitted for scribing confirmation. Total 5 samples submitted

	Particle Size	Judgement
SAMPLE A	Level 1	
SAMPLE B	Level 2	
SAMPLE C	Level 3	
SAMPLE D	Level 4	
SAMPLE E	Level 5	

NO	FINDING	PIC	COUNTERMEASURE	PLANNED DATE	EVIDENCE
1	Checksheet for CCD ranking M/C have no record on before resetting bias data. Without the data, it will be impossible to categorize error level as stated in MM-PR-OP-2180-57.	Ms Salwah	Revised checksheet for CCD ranking M/C to include before resetting bias value.	5-Sep-14	
2	a) Checking dimension of 1 pcs / rank for inspection report was not considered as good QA level inspection. Samsung request to check 10 pcs / lot for product 0603W318 only.	Ms Salwah	To set one ranking M/C as QA ranking machine to check sample taken.	11-Sep-14	
	b) Using after firing QC check data as slit depth data for inspection report was not considered sufficient. Samsung request to check on 1 pcs / lot after visual for product 0603W318 only.	Ms Salwah	To create OPL and done slit depth measurement after visual.	11-Sep-14	
3	Monitoring graph after firing for slit depth was done based on lot. Samsung request for the monitoring to be done base on mould.	Mr Azfar	To develop system for the graph to be monitored based on mould.	1-Oct-14	
4	Mould management was not clearly done.	Mr Sinna / Mr Muto	To update mould renew & service spec and put mould service and renew quantity column on mould monitoring form.	11-Sep-14	
5	To submit sample of 1000 pcs.	Mr Izumo	Sample will be with inspection report and QA outgoing data	5-Sep-14	Need to confirm with STEM

AUDIT FINDING #1

RANKING MASTER



MM-PR-RD-2180-P01-21
REV : 4 (JUNE '15)

DATE :

MACHINE NO :

RANKING MASTER THRU HOLE CHECKSHEET

TIME		L / W						ANGLE				STR				Angle Position (4.00 ~ 6.00)	
	ITEM	L1	L2	L3	W1	W2	W3	SQR 5	SQR 6	SQR 7	SQR 8	LIN1	LIN 2	LIN 3	LIN 4	Master	Product
BEFORE	BACAAN MASTER																
	BACAAN SCREEN COMPUTER																
	PERBEZAAN BACAAN																
AFTER	BACAAN SCREEN COMPUTER																
	PERBEZAAN BACAAN																
	ROLLER CLEANING																

SERIAL NO : _____
PRODUCT TYPE : _____

JUDGEMENT : OK / NG
CHECK BY : _____

THRU-HOLE BLOCK TEST : OK / NG

TIME		L / W						ANGLE				STR				Angle Position (4.00 ~ 6.00)	
	ITEM	L1	L2	L3	W1	W2	W3	SQR 5	SQR 6	SQR 7	SQR 8	LIN1	LIN 2	LIN 3	LIN 4	Master	Product
BEFORE	BACAAN MASTER																
	BACAAN SCREEN COMPUTER																
	PERBEZAAN BACAAN																
AFTER	BACAAN SCREEN COMPUTER																
	PERBEZAAN BACAAN																
	ROLLER CLEANING																

SERIAL NO : _____
PRODUCT TYPE : _____

JUDGEMENT : OK / NG
CHECK BY : _____

THRU-HOLE BLOCK TEST : OK / NG

TIME		L / W						ANGLE				STR				Angle Position (4.00 ~ 6.00)	
	ITEM	L1	L2	L3	W1	W2	W3	SQR 5	SQR 6	SQR 7	SQR 8	LIN1	LIN 2	LIN 3	LIN 4	Master	Product
BEFORE	BACAAN MASTER																
	BACAAN SCREEN COMPUTER																
	PERBEZAAN BACAAN																
AFTER	BACAAN SCREEN COMPUTER																
	PERBEZAAN BACAAN																
	ROLLER CLEANING																

SERIAL NO : _____
PRODUCT TYPE : _____

JUDGEMENT : OK / NG
CHECK BY : _____

THRU-HOLE BLOCK TEST : OK / NG

TIME		L / W						ANGLE				STR				Angle Position (4.00 ~ 6.00)	
	ITEM	L1	L2	L3	W1	W2	W3	SQR 5	SQR 6	SQR 7	SQR 8	LIN1	LIN 2	LIN 3	LIN 4	Master	Product
BEFORE	BACAAN MASTER																
	BACAAN SCREEN COMPUTER																
	PERBEZAAN BACAAN																
AFTER	BACAAN SCREEN COMPUTER																
	PERBEZAAN BACAAN																
	ROLLER CLEANING																

SERIAL NO : _____
PRODUCT TYPE : _____

JUDGEMENT : OK / NG
CHECK BY : _____

THRU-HOLE BLOCK TEST : OK / NG

TIME		L / W						ANGLE				STR				Angle Position (4.00 ~ 6.00)	
	ITEM	L1	L2	L3	W1	W2	W3	SQR 5	SQR 6	SQR 7	SQR 8	LIN1	LIN 2	LIN 3	LIN 4	Master	Product
BEFORE	BACAAN MASTER																
	BACAAN SCREEN COMPUTER																
	PERBEZAAN BACAAN																
AFTER	BACAAN SCREEN COMPUTER																
	PERBEZAAN BACAAN																
	ROLLER CLEANING																

SERIAL NO : _____
PRODUCT TYPE : _____

JUDGEMENT : OK / NG
CHECK BY : _____

THRU-HOLE BLOCK TEST : OK / NG

PANDUAN/REFERENCE :

- 1) PERBEZAAN UNTUK BACAAN MASTER DAN SCREEN COMPUTER : 0.005
- 2) MELEBIHI 0.005, MACHINE RE-SETTING SEMULA
- 3) SELEPAS RESETTING SEMULA, BACAANNYA TETAP MELEBIHI 0.005, STOP MC & BERITAHU KEPADA TEHNICIAN.
- 4) BACAAN ANGLE POSITION MASTER DAN PRODUK TIDAK BOLEH MELEBIHI 4.00~6.20

MASTER CONFIRMATION=> 10pcs

P. 1

No.	RANK	L1 W1	L2 W2	L3 W3	LX WX	LLX WWX	ANG1 STR1	ANG2 STR2	ANG3 STR3	ANG4 STR4	STR5
1.	42	60.33	60.32	60.34	60.33	0.02	-0.01	0.02	-0.01	0.02	
2.	32	49.46	49.46	49.45	49.46	0.02	-0.01	0.01	0.01	-0.01	0.01
		60.29	60.30	60.32	60.30	0.03	0.00	0.00	0.01	0.01	
3.	42	49.45	49.46	49.45	49.45	0.03	0.00	0.01	0.01	0.00	0.00
		60.32	60.33	60.35	60.33	0.03	-0.01	0.01	0.00	0.02	
4.	42	49.45	49.46	49.45	49.45	0.03	-0.01	0.01	0.01	-0.01	0.01
		60.32	60.33	60.35	60.33	0.03	-0.01	0.00	0.01	0.02	
5.	42	49.46	49.46	49.46	49.46	0.03	-0.01	0.01	0.01	0.00	0.01
		60.33	60.33	60.35	60.34	0.02	-0.01	0.02	0.00	0.02	
6.	42	49.45	49.46	49.45	49.45	0.02	-0.01	0.01	0.01	0.00	0.01
		60.32	60.33	60.34	60.33	0.02	-0.01	0.00	0.00	0.02	
7.	*****	49.46	49.45	49.45	49.45	0.01	0.00	0.00	0.00	0.02	0.00
		60.32	60.32	60.33	60.32	0.02	0.00	0.01	0.00	0.00	0.00
8.	42	49.47	49.45	49.46	49.46	0.01	0.01	0.00	0.01	0.00	0.00
		60.33	60.33	60.34	60.33	0.01	-0.01	0.00	0.00	0.01	
9.	42	49.46	49.46	49.45	49.46	0.01	0.00	0.01	0.00	-0.01	0.01
		60.32	60.33	60.34	60.33	0.02	-0.02	0.01	0.01	0.01	
10.	*****	49.45	49.46	49.45	49.45	0.02	-0.01	0.01	0.01	0.00	0.00
		60.32	60.32	60.34	60.33	0.02	-0.01	0.01	0.01	0.00	0.00
11.	42	49.45	49.47	49.46	49.46	0.02	0.00	0.01	0.01	-0.01	0.02
		60.33	60.33	60.35	60.34	0.02	0.00	0.01	0.01	0.00	
12.	42	49.44	49.46	49.45	49.45	0.02	0.00	0.00	0.02	0.00	0.01
		60.33	60.32	60.34	60.33	0.02	0.00	0.01	0.01	0.00	
13.	32	49.45	49.45	49.46	49.45	0.02	0.00	0.01	0.01	0.00	0.01
		60.31	60.31	60.31	60.31	0.00	0.02	-0.01	0.00	0.01	
14.	32	49.47	49.44	49.46	49.46	0.00	0.02	0.00	0.00	0.00	0.01
		60.30	60.31	60.32	60.31	0.02	0.03	-0.03	0.02	0.00	
15.	42	49.46	49.44	49.45	49.45	0.02	0.03	0.01	0.02	0.00	0.01
		60.33	60.32	60.34	60.33	0.02	0.00	0.01	0.00	0.01	
16.	42	49.46	49.46	49.46	49.46	0.02	0.00	0.01	0.01	-0.01	0.01
		60.32	60.33	60.34	60.33	0.02	-0.01	0.00	0.00	0.02	
17.	42	49.46	49.46	49.46	49.46	0.02	-0.01	0.01	0.01	0.00	0.00
		60.33	60.33	60.35	60.34	0.02	0.00	0.00	0.00	0.01	
18.	42	49.45	49.45	49.45	49.45	0.02	0.00	0.02	0.01	0.00	0.01
		60.32	60.33	60.34	60.33	0.02	0.00	0.00	0.01	0.00	0.01
19.	*****	49.46	49.43	49.44	49.44	0.02	0.00	0.01	0.01	0.00	0.01
		60.32	60.32	60.34	60.33	0.02	0.00	0.00	0.01		
20.	42	49.44	49.45	49.44	49.44	0.02	0.00	0.01	0.01	-0.01	0.01
		60.33	60.32	60.34	60.33	0.02	-0.01	0.01	0.00	0.01	
21.	42	49.45	49.46	49.45	49.45	0.02	0.00	0.01	0.01	-0.01	0.02
		60.32	60.33	60.35	60.33	0.03	-0.02	0.01	0.00	0.03	
22.	*****	0.00	0.00	0.00	0.00	0.00	-0.01	0.02	0.01	0.00	0.01
		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
23.	*****	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
24.	*****	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
25.	*****	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

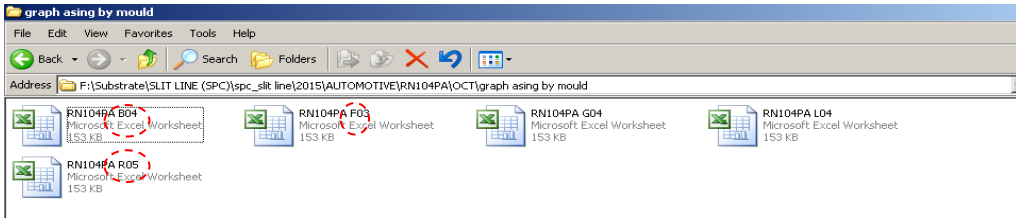
AUDIT FINDING #2B

SLIT LINE MEASUREMENT

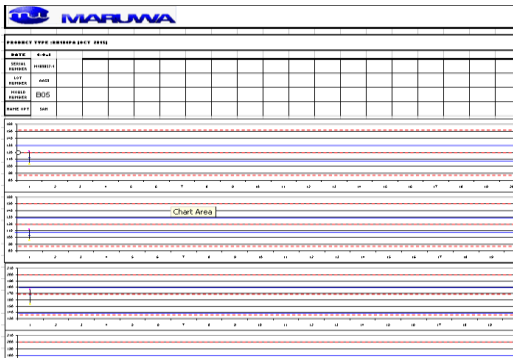
ONE POINT LESSON						CRITICAL TO QUALITY
Lesson no.	Rev. no.	Jabatan / Dept	Bahagian/Section	Proses/Kawasan	Jenis mesin	Tarikh
MM-QA-OP-2100-101	0	SUBSTRATE	QC	OUTGOING	-	16-Dec-14
☐ Tugas Asas ☐ Penyelesaian Masalah ☒ Tugas Inspection ☐ Others		Rujukan kepada Work Instruction		Prepared by	Checked by	Checked by
				DCC	Ketua Bahagian	Ketua Jabatan
DIAGRAM AND EXPLANATION				CARA-CARA PENENTUAN TITIK-TITIK PENGUKURAN SLIT LINE (GENERAL PRODUCT)		
FLOW GAMBARAJAH A				ALURAN KERJA 1) PROSES A) Cara mematahkan sample QC selepas mewarnakan produk. * Rujukan OPL: MM-QA-OP-2130B-05 (Cara-cara warnakan produk untuk Slit Line selepas Main Firing.) B) Cara-Cara penentuan titik-titik pengukuran "Slit Line" selepas "Firing". 2) FREQUENCY - 2pcs setiap lot / setiap div / Serial Number. 3) LANGKAH KERJA A) - Sample QC yang telah diwarnakan hendaklah dikeringkan terlebih dahulu. - Selepas diwarnakan produk tersebut pertulah ditandakan untuk 1 slit (Sila lihat gambarajah LD dan WD). Tujuan ditandakan adalah untuk memberitahu yang mana satu 1st line dan center. - Untuk pengukuran slit pertama haruslah ditentukan mengikut Cutting dan Index. - Bahagian yang hampir dengan cutting atau Index adalah 1a dan 1b. (rujuk gambarajah B) - Bahagian yang jauh daripada cutting daripada cutting dan Index adalah 2a dan 2b. (rujuk gambarajah B) - Cara mematahkan produk sila lihat 1 dan 2, WD dan LD seperti gambarajah A. - Produk selepas dipatah haruslah disusun mengikut susunan seperti gambarajah A. B) - Setiap pengukuran haruslah dibuat didalam garisan yang telah ditandakan pada sample tersebut. - Jumlah titik yang perlu diukur adalah 12 point bagi semua produk. WD = 12 Point (1st line 4 point, Center 8 point). LD = 12 Point (1st line 4 point, Center 8 point). - Selepas itu produk tersebut diletakkan dibawah measurescope untuk dibuat pengukuran slit line. (gambarajah C) - Data Slit Line yang telah diukur haruslah direkodkan didalam SLIT LINE SYSTEM. LANGKAH KESELAMATAN KERJA BERGILIR - GILIR UNTUK MENGELOKKAN DARIPADA SAKIT BELAKANG DAN KELETIHAN PADA MATA.		
GAMBARAJAH B PENGUKURAN SLIT LINE DI BAHAGIAN WD DAN LD Untuk bahagian WD, total = 12 Points (1st line 4 points, Center slit 8 points) diukur. Untuk bahagian LD total = 12 Points (1st line 4 points, Center slit 8 points) diukur.				GAMBARAJAH C Garisan dimana pengukuran dilakukan PRODUCT DILETAKAN DI BAWAH MEASURESCOPE DATA SLIT LINE DIREKOD DALAM SLIT LINE SYSTEM		

AUDIT FINDING #3

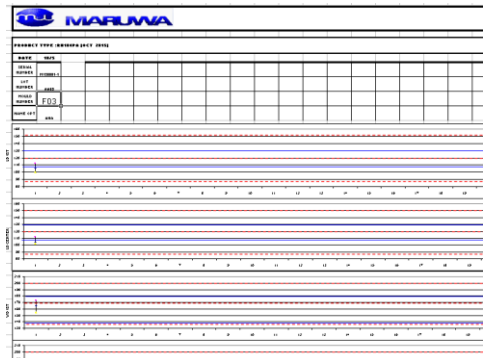
RN104PA: Slitline graph segregated by mould



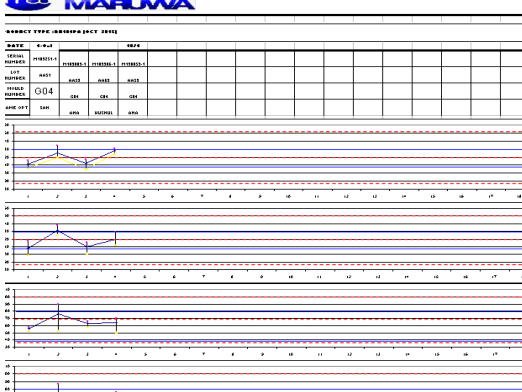
1) Mould B



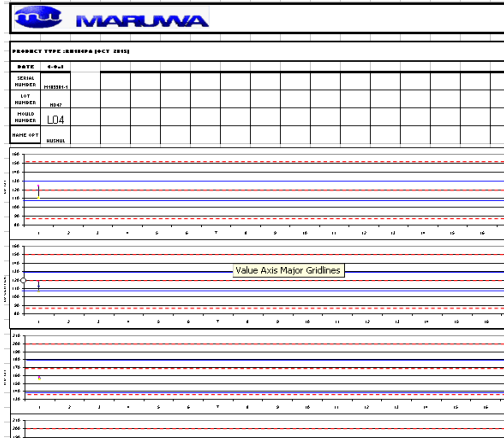
2) Mould F



3) Mould G



4) Mould L



AUDIT FINDING #4

SYSTEM CONTROL FOR PRESSING ACCUMULATE PRESS QUANTITY

Add Mould Form

MOULD INFORMATION Register Today's Date: 06 - Oct - 15

Product Name: T1276-4MB
Mould No:
Mould Type:
Press Quantity: 0
Remarks:
Washing 0 Service 0
Renew 0

Save Exit

	MouldID	DateIn	PressQty	MldType	Remarks	NonActive	washing	Service
▶	G08	13-May-15	224405	Single Moul	1	☐	40000	400000
	H01	13-May-15	1493695	Single Moul	1	☒	40000	400000
	L09	13-May-15	217271	Single Moul	1	☐	40000	400000
	D13	13-May-15	560514	Single Moul	1	☐		
	F04	07-Jul-15	108725	Single Moul		☐		
*			0			☐		

Record: 1 of 5

Automatically innactive (unable to register the mould) if exceeded renew spec (1000000)